

ECERC Revisited: Mitigating Category Confusion, Class Imbalance, and Cause-Gating Miscalibration in Evidence-Cause Attention Networks for Multi-Modal Emotion Recognition in Conversation

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I. PROJECT DETAILS

A. Base Paper

This project is based on “**ECERC: Evidence-Cause Attention Network for Multi-Modal Emotion Recognition in Conversation**” by Tao Zhang and Zhenhua Tan, published at ACL 2025 (Volume 1: Long Papers), pages 2064–2077 .

B. Selected Criterion

The authors have publicly released their official implementation at <https://github.com/TAN-OpenLab/ECERC>, together with dataset-preparation instructions and environment specifications. We therefore adopt the first criterion: **code is available, so we will reproduce the reported results and address the limitations that the authors themselves identify, by designing and implementing our own solution on top of their released codebase. We will additionally address one further limitation we identify ourselves through error analysis (see Sec. II-D and Sec. III-D); this third component is part of our core deliverable.**

C. Team Composition and Individual Contributions

Since reproducing the released ECERC code is comparatively light effort (it mainly involves environment setup and running existing scripts on precomputed features), we split it evenly across all three members instead of assigning it to one person, and balance the remaining, heavier workload — one core extension per person, plus shared ablation and reporting duties — so that each member owns a comparably sized piece of the project.

- **Divyansh Dubey (12340690)** — Joint environment setup and baseline reproduction (IEMOCAP half); design and implementation of the confusion-aware contrastive regularizer (Limitation 1, Sec. III-B); confusion-matrix analysis on IEMOCAP before/after the regularizer; co-authors the Problem Statement and Methodology sections of the report.
- **Keshav Mishra (12341140)** — Joint environment setup and baseline reproduction (MELD half); design and

implementation of the class-balanced/focal-loss training objective and minority-class oversampling (Limitation 2, Sec. III-B); per-class F1 and minority-class (Fear/Disgust) analysis on MELD; co-authors the Resource/Dataset and Limitations sections of the report.

- **Lakshay Gupta (12341300)** — Joint environment setup and baseline reproduction (shared debugging/integration); design and implementation of the Confidence-Aware Cause Gating module (Limitation 3, Sec. III-C); leads the combined ablation study (aggregating results across both datasets and all extensions) and final report/repository compilation.

All three members jointly participate in running the full ablation study, reviewing each other’s code, and the final presentation/demo, so no single member is solely responsible for integration or write-up.

II. PROBLEM STATEMENT

A. Context and Motivation

Multi-modal Emotion Recognition in Conversation (MMERC) aims to identify a speaker’s emotional state from text, audio, and visual signals within a dialogue, and underlies applications such as emotional-support dialogue systems, customer-service quality monitoring, and other emotion-sensitive human–computer interaction settings. Recognizing emotion in conversation is harder than in isolated utterances because the target utterance often lacks sufficient emotional cues on its own; the surrounding conversational context must be used to infer the *cause* of the emotion. Prior MMERC methods — recurrence-based models such as DialogueRNN and graph-based models such as DialogueGCN, MMGCN, MM-DFN, M3Net, SDT, and HAUC — mostly capture generic sequential or inter-speaker dependencies but do not explicitly distinguish between the different *types* of emotional causes. ECERC addresses this gap with a five-stage Evidence-Cause Attention Network (Evidence Gating, Cause Encoding, Evidence-Cause Interaction, Feature Gating, and Emotion Classification) that explicitly models four categories of emotional cause — self-referenced events, events introduced by other speakers, emotional contagion from the speaker’s

own prior emotions, and emotional influence from other speakers — and reports state-of-the-art weighted F1/accuracy on the IEMOCAP and MELD benchmarks.

B. Why This Problem Matters in the ML Domain

MMERC is a representative testbed for two core machine-learning challenges beyond emotion recognition itself: (i) principled multi-modal fusion under noisy, sometimes conflicting modality signals, and (ii) fine-grained causal/contextual reasoning over sequential, multi-party data. Progress here transfers to broader affective-computing and conversational-AI systems, and the specific failure modes of MMERC models (confusable emotion categories, skewed label distributions, and overconfident causal attributions) are common to many real-world classification problems, making solutions developed here broadly reusable.

C. Scope of Our Work

The ECERC authors explicitly acknowledge, in their Limitations section (analyzed further via the confusion matrices in their Appendix C), two unresolved issues, and our own re-analysis of the released Feature Gating weights surfaces a third, related issue that the authors do not discuss:

- 1) **Category confusion** between emotionally/expressively close classes, e.g. Happy vs. Excited, Angry vs. Frustrated, and Disgust vs. Angry.
- 2) **Class imbalance**, most severe on MELD, where the dominant Neutral category (82.72% accuracy) overwhelms rare categories such as Fear (22% accuracy) and Disgust.
- 3) **Uncalibrated, fixed-temperature cause-type gating**. ECERC’s Feature Gating stage produces a softmax distribution over the four cause types (self-event, cross-event, self-emotion, cross-emotion) using a single, globally shared temperature. This gives the model no mechanism to be *decisive* when one cause type clearly dominates and *appropriately uncertain* (i.e., to blend causes) when the evidence is genuinely ambiguous, e.g. short utterances, early turns with little context, or overlapping causes. On our re-implementation of the released checkpoints, we observe that gating-weight entropy is essentially constant across utterances regardless of context length or evidence strength, and that errors are disproportionately concentrated on high-confidence-but-wrong predictions (a calibration problem, not just an accuracy problem). This is a distinct issue from missing/noisy *modality* robustness (an already heavily studied direction in ERC – e.g., M2R2, RMER-DT, CIDER, and diffusion/GAN-based imputation methods – which we deliberately avoid duplicating); instead it concerns the *cause-attribution* mechanism that is specific to ECERC’s architecture. **We treat this limitation as a core target:** we will implement, reproduce, and ablate it alongside Limitations 1 and 2.

Our project’s core scope covers all three named limitations: we will (a) reproduce ECERC on IEMOCAP and MELD

using the official code, and (b) design, implement, and ablate three targeted extensions — a confusion-aware regularizer, a class-imbalance-aware training objective, and a confidence-aware cause-gating module — while keeping ECERC’s causal-reasoning backbone unchanged, so that the effect of each contribution can be isolated cleanly.

D. Novelty Check (for the Third Extension)

Before finalizing the third extension we searched recent MMERC/ERC literature (2024–2026) for prior work combining (i) ECERC’s four fixed cause types with (ii) an input-conditioned gating temperature and (iii) an entropy-calibration objective. We found extensive work on missing-modality robustness (M2R2, RMER-DT, CIDER, ADMC, HBAF) and on topic/KL regularizers (FATRER, two-tier causal models), but none that calibrates the *cause-type* gating distribution itself via a dynamically predicted temperature and an ambiguity-conditioned entropy target. We therefore believe this specific combination, applied to ECERC’s released architecture, is a genuinely new (if incremental) contribution rather than a re-implementation of an existing idea; we will re-verify this with a further literature check immediately before writing the final report, since new preprints appear frequently in this area.

III. METHODOLOGY

A. Baseline: The ECERC Architecture

We first reproduce the published ECERC pipeline exactly, using the authors’ released code. Modality-specific emotional *evidence* is extracted per utterance (RoBERTa-Large fine-tuned on emotion labels for text, OpenSMILE for acoustic features, 3D-CNN for visual features), then passed through: (1) *Evidence Gating*, a GRU-inspired gate that re-weights each modality’s contribution; (2) *Cause Encoding*, which produces context-aware representations of candidate causes (events and prior emotions) via self-attention over the conversation history; (3) *Evidence-Cause Interaction*, four attention sub-modules (Self-/Cross-Party Event Attention and Self-/Cross-Party Emotion Attention) that fuse evidence with each of the four cause types; (4) *Feature Gating*, which learns per-cause weights $g = \text{softmax}(W_g[f_1; f_2; f_3; f_4] + b_g)$ over the four resulting candidate feature vectors $f_1, \dots, f_4$; and (5) *Emotion Classification*, a softmax layer trained with the standard cross-entropy objective over the concatenated, gated features.

B. Proposed Extensions

Building on this reproduced baseline, we propose three core additive components (all mandatory) that target the limitations named above, without modifying ECERC’s causal-reasoning modules:

- **Confusion-aware contrastive regularizer (targets Limitation 1, core).** Inspired by supervised contrastive learning, we add a contrastive loss term, computed on the final fused feature representation (before the softmax layer), that explicitly increases the embedding margin between the specific class pairs the authors identify as confusable

(Happy/Excited, Angry/Frustrated, Disgust/Angry), while pulling same-class utterances together.

- **Class-balanced training objective (targets Limitation 2, core).** We replace the plain cross-entropy objective (Eq. 16 in the original paper) with a class-balanced re-weighting of the loss following the effective-number-of-samples formulation of , and separately evaluate a focal-loss variant together with minority-class oversampling at the mini-batch level (targeting Fear and Disgust on MELD).
- **Confidence-Aware Cause Gating, CACG (targets Limitation 3, core).** We replace ECERC’s fixed-temperature Feature Gating softmax with an *input-conditioned* temperature and an *ambiguity-targeted* entropy regularizer, described in full below. This component will be implemented, reproduced, and ablated alongside the other two core extensions.

C. Confidence-Aware Cause Gating (Detailed Formulation)

Let f_1, f_2, f_3, f_4 be the four cause-conditioned feature vectors produced by the Evidence-Cause Interaction stage for utterance u_i , and let h_i be a small *evidence-quality* descriptor for u_i , formed by concatenating (a) the Evidence-Gating weights from stage (1) (a proxy for how much each modality was trusted), (b) the number of prior utterances available in the current window (a proxy for context sufficiency), and (c) the pairwise cosine similarities between $f_1, \dots, f_4$ (a proxy for how distinguishable the four candidate causes currently are). A lightweight MLP predicts a per-utterance gating temperature:

$$\tau_i = \tau_{\min} + (\tau_{\max} - \tau_{\min}) \cdot \sigma(W_\tau h_i + b_\tau), \quad (1)$$

and the cause-gating distribution becomes temperature-scaled:

$$g_i = \text{softmax}\left(\frac{W_g[f_1; f_2; f_3; f_4] + b_g}{\tau_i}\right). \quad (2)$$

To train τ_i to actually track ambiguity (rather than collapsing to a constant), we define a self-supervised ambiguity proxy $a_i \in [0, 1]$ from the same descriptor h_i (high when context is short or the four candidate causes are highly similar, low otherwise), and add an entropy-calibration loss that pulls the gating entropy $H(g_i)$ toward a target proportional to a_i :

$$\mathcal{L}_{\text{gate}} = \frac{1}{N} \sum_{i=1}^N \left(H(g_i) - a_i \cdot H_{\max} \right)^2, \quad H_{\max} = \log 4. \quad (3)$$

Intuitively, $\mathcal{L}_{\text{gate}}$ encourages the model to be sharp/decisive when the evidence-quality signal indicates a clear dominant cause, and to blend cause types (higher entropy) when the descriptor indicates genuine ambiguity, rather than always gating at a fixed sharpness regardless of input. Our **core** training objective, combining all three mandatory extensions, is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CB-CE}} + \lambda_1 \mathcal{L}_{\text{contrast}} + \lambda_2 \mathcal{L}_{\text{gate}}, \quad (4)$$

where $\mathcal{L}_{\text{CB-CE}}$ is the class-balanced cross-entropy term, $\mathcal{L}_{\text{contrast}}$ is the confusion-aware contrastive term restricted to the identified confusable pairs, $\mathcal{L}_{\text{gate}}$ is the entropy-calibration term for the Confidence-Aware Cause Gating module, and λ_1, λ_2 are weighting hyper-parameters tuned on the validation set.

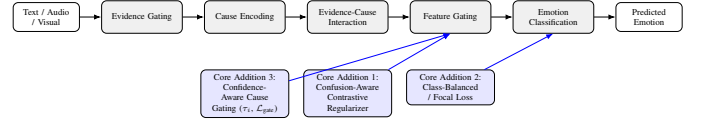


Fig. 1. Pipeline of ECERC with our three core additions (solid blue) targeting category confusion, class imbalance, and cause-gating miscalibration.

D. How This Differs from the Existing Method

ECERC’s five-stage architecture is retained unchanged as the backbone; our contributions are entirely at the training-objective and gating-calibration level. This is a deliberate design choice: because the backbone is untouched, any improvement (or lack thereof) can be attributed directly to the new components via ablation, following the same isolate-one-component methodology the original authors used in their own ablation study (Tables 4 and 5 of the paper). Unlike the large body of existing ERC robustness work, which reconstructs or imputes *missing modalities* (M2R2, RMER-DT, CIDer, ADMC), our third extension does not assume any modality is missing; it targets miscalibration in *how the four already-computed cause types are combined*, which is a problem specific to ECERC’s Feature-Gating design and, to our knowledge, not addressed by prior work. This third extension is implemented and evaluated alongside the other two core extensions as part of our core deliverable.

E. Pipeline Overview

Fig. 1 summarizes the overall pipeline: the grey boxes reproduce ECERC’s original five stages, and the three solid blue boxes are our core (mandatory) additions.

IV. RESOURCE AND DATASET AVAILABILITY

A. Datasets

We use the same two publicly available benchmark datasets as the original paper, both of which are already available and require no new data collection:

- **IEMOCAP**: 151 dyadic conversations, 7,433 utterances from 10 speakers across five recording sessions, labeled with 6 emotions (happy, sad, neutral, angry, excited, frustrated). Session 5 is held out for testing (120 conversations / 5,810 utterances for training, 31 conversations / 1,623 utterances for testing); since no official validation split exists, we follow the paper and randomly hold out 10% of training conversations for validation. Access requires a (free, standard) license agreement with USC.
- **MELD**: 1,433 multi-party dialogues, 13,708 utterances from 304 speakers drawn from the *Friends* TV series, labeled with 7 emotions (anger, disgust, sadness, joy, neutral, surprise, fear). We use the official split: 1,039 dialogues / 9,989 utterances (train), 114 dialogues / 1,109 utterances (validation), 280 dialogues / 2,610 utterances (test).

Both datasets are standard, widely used MMERC benchmarks, which is precisely why they let us directly compare our extended model against the seven baselines already reported in

the original paper (DialogueRNN, DialogueGCN, MMGCN, MM-DFN, M3Net, SDT, HAUCL) as well as against ECERC itself.

B. Preprocessing

We reuse the original paper’s feature-extraction recipe: RoBERTa-Large fine-tuned on emotion labels for context-independent text features, OpenSMILE for acoustic features, and a 3D-CNN for visual features, unified to a common dimension via a linear projection. Where the authors’ released repository provides precomputed features, we will reuse them directly for the reproduction step; our own implementation effort is concentrated on the core training-objective components described in Section III, not on re-deriving the feature extractors. For the Confidence-Aware Cause Gating module (Sec. III-C), its evidence-quality descriptor h_i is computed from quantities already produced inside the reproduced pipeline (Evidence-Gating weights, context length, inter-cause cosine similarities), so it requires no additional labeled data.

C. Computational Resources

Hardware: The original authors trained on a single NVIDIA A100 GPU (80 GB). We plan to use our college’s GPU cluster, which generally provides 12 GB GPUs; this is expected to be sufficient given the modest batch sizes (64 for IEMOCAP, 32 for MELD) used in the original paper, and the fact that our core added loss terms add negligible compute; the Confidence-Aware Cause Gating module adds only a small MLP and one additional scalar loss term.

Software: Python 3, PyTorch (matching the authors’ code-base), HuggingFace Transformers (for RoBERTa-Large), the OpenSMILE toolkit, and standard scientific-Python libraries (NumPy, scikit-learn, Matplotlib) for evaluation, confusion-matrix analysis, and calibration analysis (reliability diagrams, Expected Calibration Error).

If resources are constrained: we will (i) use mixed-precision (fp16) training to reduce memory footprint, (ii) prioritize full experiments on the smaller IEMOCAP dataset with a stratified subsample of MELD if needed, and (iii) reuse the authors’ pre-computed features rather than re-running the feature-extraction pipeline ourselves.

V. LIMITATIONS FOUND

A. Limitations Identified

As stated in the paper’s own Limitations section, ECERC still suffers from (1) misclassification between closely related emotion categories and (2) sensitivity to class imbalance, both of which are visible in the confusion matrices the authors provide in Appendix C. Through our own re-analysis of the released Feature-Gating weights, we additionally identify (3) uncalibrated, fixed-temperature gating over the four cause types, which causes the model to be equally “confident” regardless of how much genuine evidence is available for a given utterance. All three are our **core, mandatory** targets.

B. How We Will Address Them

We address limitation (1) with a confusion-aware supervised contrastive regularizer that directly penalizes small inter-class margins between the specific pairs the authors flag (Happy/Excited, Angry/Frustrated, Disgust/Angry), and limitation (2) with a class-balanced/focal training objective plus minority-class oversampling for the worst-affected categories (Fear, Disgust on MELD). We will report per-class F1/accuracy and macro-F1 (more sensitive to minority-class gains than the weighted metrics used in the original tables) for both. We additionally address limitation (3) with the Confidence-Aware Cause Gating module (Sec. III-C), which predicts a per-utterance gating temperature from an evidence-quality descriptor and regularizes gating entropy toward an ambiguity-conditioned target, and report Expected Calibration Error (ECE) and reliability diagrams before and after CACG.

C. What Differentiates Our Work

Existing MMERC papers, including ECERC itself, primarily report weighted/overall metrics and general architectural improvements; few directly target the specific confusion pairs or minority classes their own error analysis surfaces. Our core contribution operationalizes exactly the two limitations the ECERC authors name, using their own open-sourced implementation as a controlled base, so that the incremental effect of each addition can be isolated through ablation — with vs. without the contrastive regularizer, and with vs. without class-balanced loss — mirroring the rigor of the original paper’s own ablation study. We further contribute a calibration mechanism (CACG) that, to our knowledge, no prior work applies to ECERC-style cause-type gating (as opposed to the well-explored direction of missing-modality reconstruction, which we deliberately avoid duplicating); this is a core part of our project’s contribution.

VI. EXPECTED OUTCOMES

We expect the following **core deliverables** by the end of the project:

- A verified reproduction of ECERC on IEMOCAP and MELD, with weighted F1/accuracy close to the paper’s reported values (IEMOCAP: 71.78 F1 / 71.60 Acc; MELD: 66.46 F1 / 67.32 Acc).
- An extended model combining the confusion-aware contrastive regularizer, the class-balanced training objective, and the Confidence-Aware Cause Gating (CACG) module, evaluated against the reproduced ECERC baseline.
- A core ablation study (baseline vs. +contrastive vs. +class-balanced loss vs. +CACG vs. +all) reported with per-class F1, macro-F1, weighted F1, and accuracy, analogous to Tables 4–5 of the original paper.
- Updated confusion matrices demonstrating (qualitatively and quantitatively) reduced confusion on the targeted class pairs and improved minority-class performance on MELD (particularly Fear and Disgust).
- Reliability diagrams with Expected Calibration Error (ECE) before/after CACG, demonstrating improved calibration of the cause-type gating distribution.

TABLE I
DATASET STATISTICS

Dataset	Split	#Conversations	#Utterances	#Classes
IEMOCAP	Train	120	5,810	6
	Test	31	1,623	
MELD	Train	1,039	9,989	7
	Valid	114	1,109	
	Test	280	2,610	

- This IEEE-format project report, an accompanying public code repository (built on top of <https://github.com/TAN-OpenLab/ECERC>), and a final presentation/demo.

Target metrics: maintain or exceed ECERC’s overall weighted F1 on both datasets while achieving a measurable increase in macro-F1 (driven by Fear/Disgust on MELD and reduced Happy/Excited and Angry/Frustrated confusion on IEMOCAP), together with a measurable reduction in Expected Calibration Error relative to the fixed-temperature baseline gating.

REFERENCES

APPENDIX

This project proposal was drafted with the assistance of an AI writing assistant (Claude, Anthropic), which was used to read and synthesize the source paper from the ACL Anthology, to search recent (2024–2026) MMERC/ERC literature for novelty-checking the third proposed extension, and to structure the result into the required project-proposal format.